

DEEPAK KUMAR

Birla Institute of Technology, Mesra, Ranchi

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SUMMARY

GenAI Backend Developer and Software Engineer with strong foundations in **Mathematics and Computing**. Proven track record of architecting scalable **MERN stack** applications, high-performance **Desktop Tools** using **Python/OpenCV**, and **Retrieval-Augmented Generation (RAG)** pipelines. Experienced in designing secure APIs (**JWT**, **OAuth 2.0**) and integrating LLMs using **LangChain** and **FastAPI**. Seeking to contribute AI-driven backend solutions to innovative engineering teams.

EDUCATION

Birla Institute of Technology, Mesra

May 2027 (Expected)

Integrated M.Sc. in Mathematics and Computing | CGPA: 8.71/10

Ranchi, Jharkhand

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Computer Networks.

TECHNICAL SKILLS

Languages: C++, Java, JavaScript (ES6+), Python, SQL

Backend & AI: Node.js, Express.js, FastAPI, LangChain, LangGraph, ChromaDB, MySQL, MongoDB

Frontend & Libraries: React.js, Tailwind CSS, Streamlit, Pydantic, PyPDF, Cheerio, Redux

DevOps & Tools: Git, GitHub Actions, Docker, Postman, Uvicorn, Python-dotenv, JWT

EXPERIENCE

Full Stack Developer (Freelance)

Oct 2023 – Dec 2023

LooksLove Salon

Remote | [\[Live\]](#)

- Delivered a production-ready booking system with secure **OTP-based login**, scaling to **500+ active users** and increasing digital appointment scheduling by **20%**.
- Built a comprehensive **Admin Dashboard** (React/Node.js) for real-time inventory tracking and staff management, reducing daily administrative overhead by 30%.

PROJECTS

LectureLens AI | *Python, OpenCV, NumPy, Tkinter*

[\[Live\]](#)

- Engineered a desktop video processing tool using **OpenCV** that removes **95%** of visual obstructions from lectures via a custom Progressive Background Building algorithm.
- Optimized rendering efficiency by **40%** using a **Dual-Resolution strategy** (processing low-res grayscale for motion analysis), generating 1080p PDF notes without quality loss.

TnP Notifier | *Node.js, GitHub Actions, Cheerio, MongoDB*

[\[GitHub Link\]](#)

- Developed an automated placement alert system serving **200+ students**, eliminating manual monitoring and saving approximately **15 hours** of collective student time weekly.
- Orchestrated self-healing **GitHub Actions** CRON workflows with **99.9% uptime**, ensuring zero missed notifications for critical job announcements.

GenAI Chatbot Portfolio | *LangChain, FastAPI, ChromaDB, React.js*

[\[Live\]](#)

- Architected a Retrieval-Augmented Generation (RAG) agent using **LangChain**, **Gemini API**, and **ChromaDB** to embed, parse, and interactively query PDF resume data.
- Developed a robust backend API utilizing **FastAPI**, **Uvicorn**, and **Pydantic** for data validation, seamlessly integrating the conversational agent into a React portfolio application.

PUBLICATIONS

OAuth 2.0 Security Analysis | *Paper ID: 328 (Under Review)*

- * Authored a research paper analyzing authorization code flow vulnerabilities; proposed a token integrity protocol to mitigate **CSRF** and **Replay Attacks**.

ACHIEVEMENTS

- **Problem Solving:** Solved **600+ DSA questions** across various coding platforms (LeetCode, GFG).
- **Competitive Programming:** Secured **1668 Global Rank** in LeetCode Weekly Contest 504.
- **IEEE Conference Presenter (2025):** Presented poster on "OAuth 2.0 in Healthcare" at NIT Jalandhar. [\[Certificate\]](#)
- **Startup Founder (2022-2023):** Founded **Earthen Pots**, leading a team of 2 to manage production and D2C operations.
- **Volunteer (2022):** International Mathematics Conference, assisting in event organization and logistics.
- **NCC 'C' Certificate:** Demonstrated discipline, physical fitness, and leadership in annual training camps.